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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.C.A. (CBCS) Degree Examinations

May - June 2024

22MCA222 – HEALTH CARE ANALYTICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each unit.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Discuss on several modalities that are available in medical imaging with suitable instances.	10	L*1	1	2
	b)	Explain the significance of analytics in healthcare management and some of its usages.	10	L2	1	1
2.	a)	What components constitute healthcare analytics? Give examples to illustrate.	10	L1	1	2
	b)	Define healthcare analytics. Discuss its significance in the process of converting data to knowledge and in healthcare industry.	10	L2	1	1
		Unit – II				
3.	a)	List and explain several types of biological signals.	10	L2	2	1
	b)	What difficulties do biological signal analysis professionals face in the field of healthcare? Discuss each challenge in detail.	10	L2	2	2
4.	a)	Explain how sensor data mining is used in healthcare industry with suitable examples.	10	L2	2	1
	b)	Discuss the significance of analytics in improving patient's outcome in healthcare and with suitable examples.	10	L2	2	2
		Unit – III				
5.	a)	Discuss the various types of data integration, emphasizing the benefits and limitations.	10	L2	3	2
	b)	Define genomic data. Explain various key steps and methods involved in genomic data analysis.	10	L2	3	1
6.	a)	Elaborate the workflow of standard bioinformatics to analyze the genomic data with neat diagram.	10	L2	3	2
	b)	How Clinic genomic Integration works? Discuss the issues and challenges of integrating genomic and clinical data.	10	L2	3	1
		Unit – IV				
7.	a)	What are electronic health records? Enumerate and describe the components of EHR.	10	L2	4	1
	b)	Discuss the challenges and applications of processing medical information.	10	L2	4	2
8.	a)	Explain about how social media may be used to analyse, identify, and monitor infectious disease epidemics.	10	L2	4	1
	b)	Demonstrate the concept of natural language processing in the field of healthcare, highlighting its components and overall process.	10	L2	4	2
		Unit – V				
9.	a)	Discuss the benefit of visual analytics in healthcare and how it can be used by patients, physicians, and clinical process.	10	L2	5	2
	b)	Discuss the following:				
		i) Cross validation ii) ROC curve iii) Brier score	10	L2	5	2
		iv) Cost sensitive learning v) Sequential Pattern Mining				
10.	a)	Explain the various models of statistical prediction that are utilized in data analytics for healthcare.	10	L2	5	2
	b)	Describe the popular temporal pattern mining methods.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
